

New way to 3D print graphene

Researchers from Virginia Tech University have found a way to 3D print graphene. <https://dgit.in/sep18-31-1>



AI gives you the moves

Researchers can copy the movements of a subject's body in one video onto another person's body in another video. <https://dgit.in/sep18-31-2>

tech

DIY

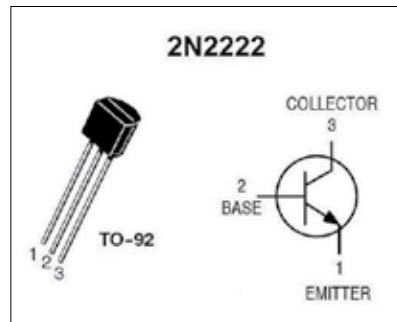
the primary coil - using the thicker magnet wire or even a single strand insulated copper wire. We'll wind it on the same tube but this time, the direction of the winding will be the opposite of the previous coil. This is an important step. If you wound the 30 AWG coil clockwise, then the thicker and shorter wire needs to be wound counter-clockwise.

Now, you have to decide the number of turns. The way transformer coils work, we're trying to reach a certain ratio between the 30 AWG wire and the 20 AWG wire. In order to amplify the voltage, the factor which determines the magnitude of the amplification is the ratio that we spoke of. Since our source is a 9 Volt battery, we're trying to amplify it by a factor of 100. So if the primary has three turns, then the secondary needs to have around 300 turns. This is why we used a thinner wire for the secondary, however, you can use thicker wire too as long as you manage to get 300 turns on it.

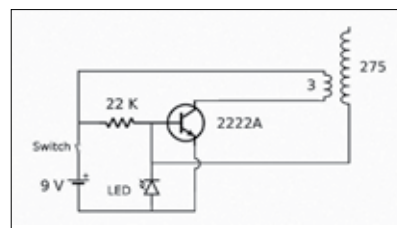
STEP 2 - MAKING THE CIRCUIT

The circuit is fairly simple as we've mentioned previously. The NPN Bipolar Junction Transistor sits at the heart of the circuit. For the uninitiated, the circular symbol with the tiny arrow on the inside is the 2N2222 transistor. The arrow indicates the Emitter contact, the fat vertical line is the base contact and the third one is the collector contact. On your average 2N2222, the emitter is pin 1, base is pin 2 and collector is pin 3. Connect the negative terminal of the battery and the anode of the LED to the emitter pin of the transistor. Then connect the second-

ary coil, one end of the resistor and the cathode of the LED to the base pin of the transistor. Connect the positive terminal of the battery to the switch in series and connect that to the other end of the resistor. Finally, connect one end of the primary coil to resistor and the other to the collector pin. The circuit diagram is below.



2N2222 pinout



Simple Slayer Exciter circuit

The value of the resistor needs to only be approximately 22K, within a tolerance of 1-2K ohms.

STEP 3 - THE MOMENT OF TRUTH

Get your CFL tube ready and throw the switch on. If the LED lights up, then your circuit should be working fine. Bring the LED tube closer to the secondary coil and it should light up. Voila! You have achieved wireless power transmission.

In all likelihood, your first attempt did not work. Which is probably why you're reading this line. Well, worry not, ours didn't work the first time either. Grab your multimeter and check all the components and check for continuity between the components. There's a chance that you might have switched the polarity of the primary coil. Swap the connections and see if it works again. Also, ensure that you are using the right transistor, if the Vce and hfe values are out of spec, then you will probably see the LED light up but the CFL will not light up.

WHAT'S HAPPENING HERE?

When you turn the power on, the resistor drives the base of the NPN transistor, thus, switching it on. The transistor then powers the primary coil of the transformer. A magnetic field is generated by the primary. This field induces power into the secondary coil. Since the ratio of the turns on the primary to the secondary is approximately 100, there is 900 Volts running across the secondary. Because of the capacitance due to the aluminium foil ball at the end, the rising power in the secondary is opposed by it. This causes the voltage on the other end of the transformer to drop. When this happens, the base of the transistor drops and as a result the entire field collapses. Once the field collapses, the base voltage starts to rise and the entire cycle repeats. So you are generating a field that oscillates at a high frequency

You can check out a simulation of the circuit here - <https://dgit.in/SECWorking>.

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